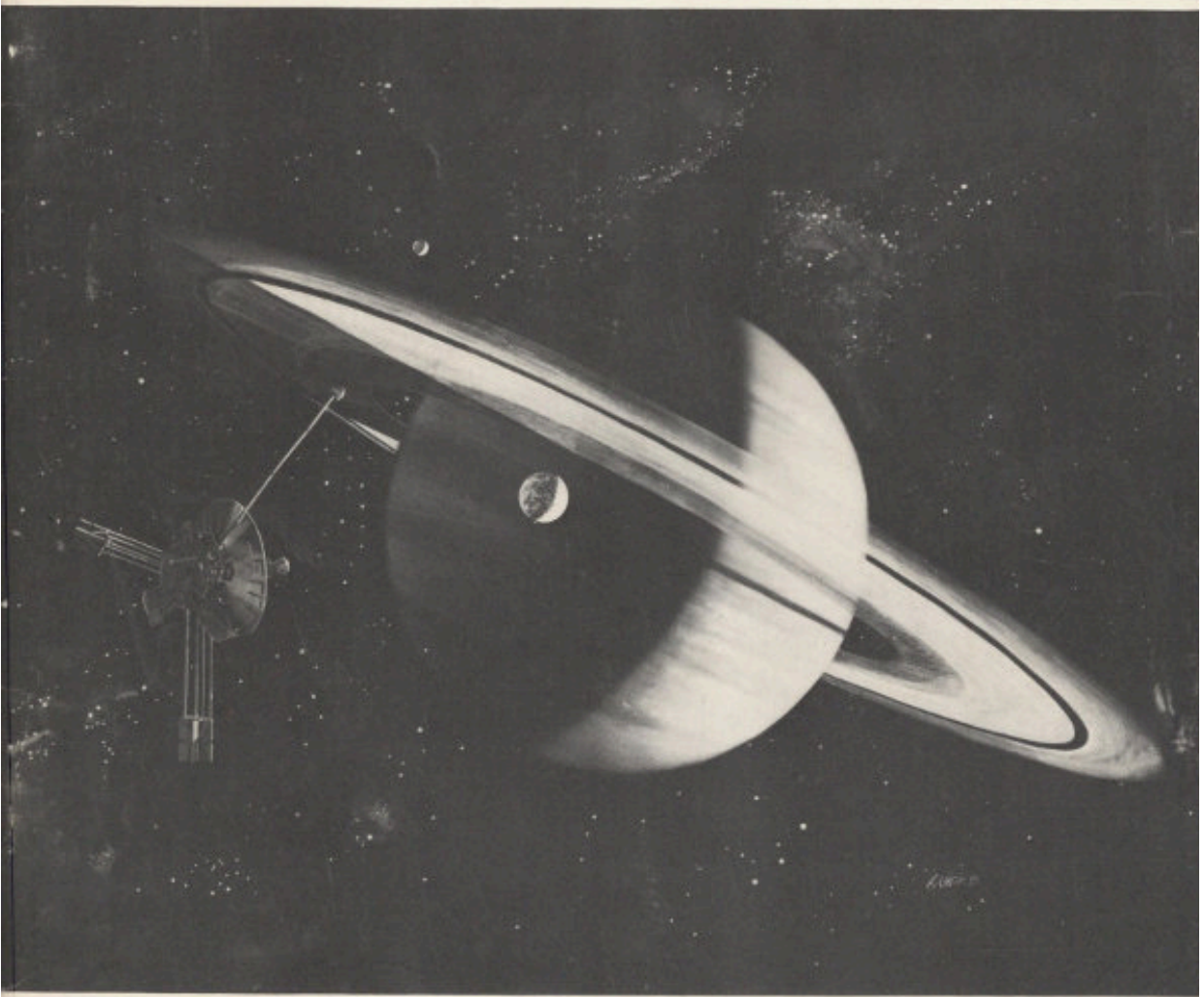


PIONEER SATURN ENCOUNTER



NASA

PIONEER SATURN ENCOUNTER



NASA

The Project Gutenberg eBook of Pioneer Saturn Encounter

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Pioneer Saturn Encounter

Author: United States. National Aeronautics and Space Administration

Release date: October 7, 2017 [eBook #55695]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/55695

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK PIONEER
SATURN ENCOUNTER ***

PIONEER SATURN ENCOUNTER



CONTENTS

Foreword	1
Introduction	3
Pioneer Saturn	4
Saturn Images	8
Science Highlights	21
Outward Bound	28

FOREWORD

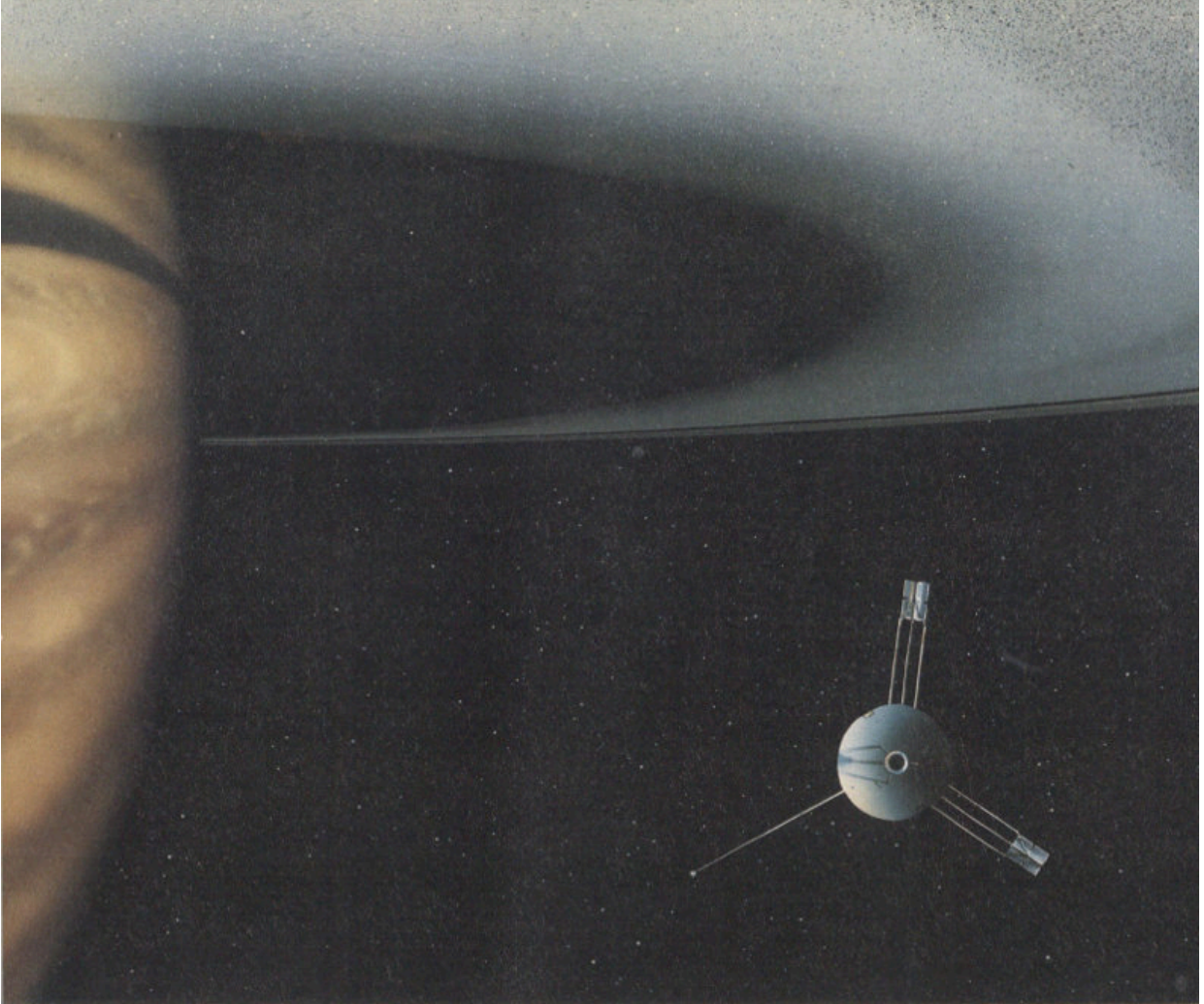
The Pioneer 10 and 11 spacecraft, launched in 1972 and 1973, respectively, were well named: they made the first crossings of the asteroid belt and were the first to encounter Jupiter and its intense radiation belts. Pioneer 11's trajectory, bent into a hairpin curve by Jupiter's powerful gravitational field, allowed it to recross the solar system to make the first flyby of Saturn almost a billion miles from Earth where it came within 13,300 miles of the cloud tops.

Assembled in this publication is a selection of the pictures returned by Pioneer 11 of Saturn and its largest moon, Titan. These images are of great beauty as well as of great scientific interest, serving to whet our appetite for the more detailed observations to be made by Voyager in 1980 and 1981. Tracking of both Pioneers will continue for many more years, providing fundamental data on the nature of interplanetary space in the depths of the solar system. The results of these outer-planet Pioneer missions have far exceeded our hopes and expectations of a decade ago when the program was initiated.

Robert A. Frosch, Administrator
National Aeronautics and Space Administration

September 1979

NASA
National Aeronautics and Space Administration
Ames Research Center
Moffett Field, California 94035



Artist's drawing of Saturn and its rings showing the Pioneer Saturn spacecraft passing under the rings and nearing closest approach to the planet. Actual pictures could not be obtained at this time because of the high speed of the spacecraft.

INTRODUCTION

We have entered into a new era of space exploration. Missions undertaken during the lunar exploration of the 1960's typically lasted a matter of days with commands issued and carried out in near real time. Now, a decade later, planetary voyages may last for many years as the spiraling trajectories of the spacecraft make periodic intersections with the orbits of the planets. Communicating with us across the vastness of space, these spacecraft report to us their experiences as they traverse the outer reaches of the solar system.

Among these deep space travelers, Pioneers 10 and 11 are appropriately named, for they truly are pioneering the exploration of the outer solar system. Launched in 1972 and in 1973, respectively, they were the first spacecraft to fly by Jupiter (in 1973 and 1974). At Jupiter, Pioneer 11's trajectory was carefully targeted to swing it toward Saturn for an encounter in September 1979. We see some of the early results in this publication.

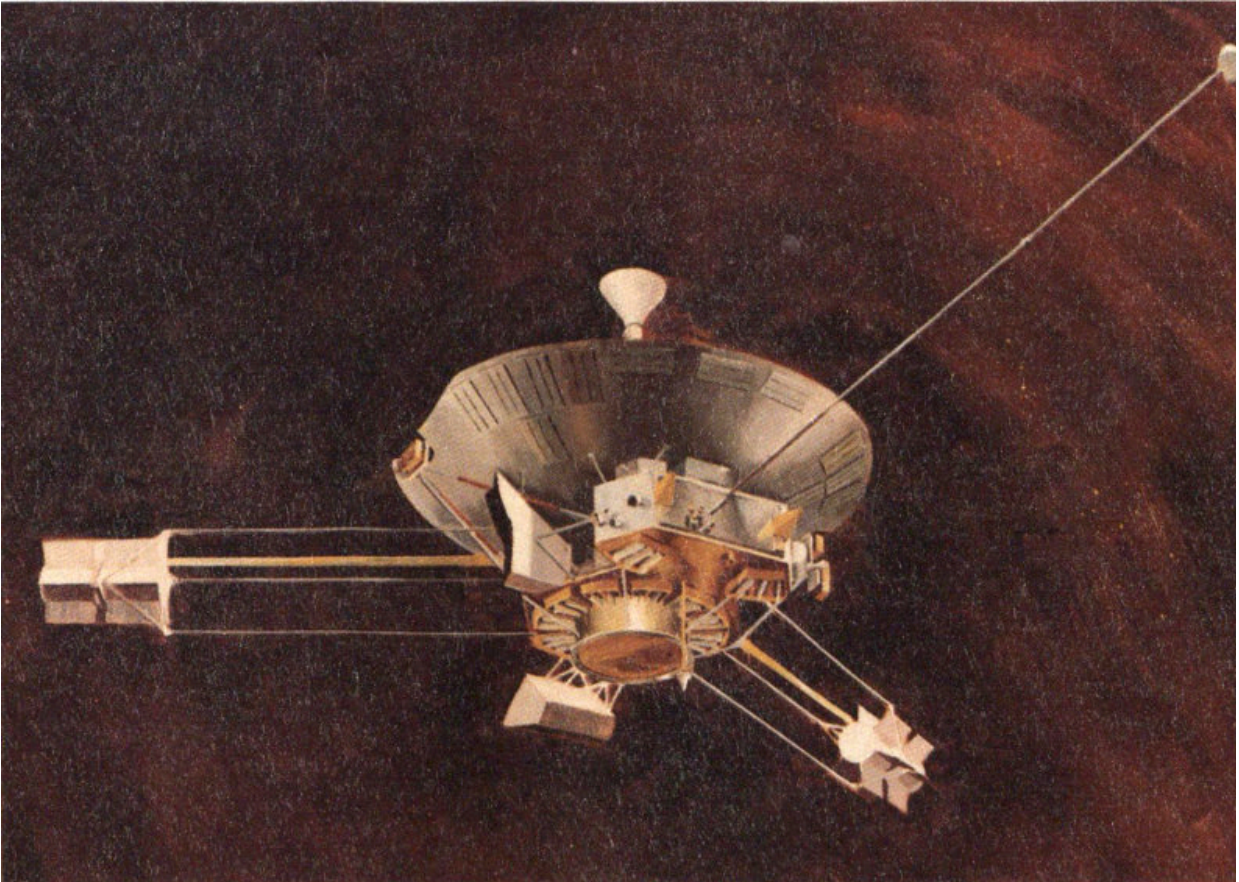
Other spacecraft are following along the trail blazed by Pioneer Saturn. Voyager 1 passed by Jupiter in March 1979 and will reach Saturn in November 1980. Voyager 2 has also passed beyond Jupiter and will encounter Saturn in August 1981, with the further possibility of traveling on to Uranus (a 1986 encounter). Under development are the Galileo orbiter and atmospheric entry probe, destined to journey to Jupiter where the orbiter will return more detailed information, including high-resolution pictures of the Galilean satellites, and the probe will penetrate deep below the Jovian clouds.

In the coming years, each of these follow-on missions will enrich our understanding of the solar system, greatly supplementing the observations of Pioneers 10 and 11. But one thing will never change. The Pioneers were first.

Thomas A. Mutch
Associate Administrator for Space Science

National Aeronautics and Space Administration

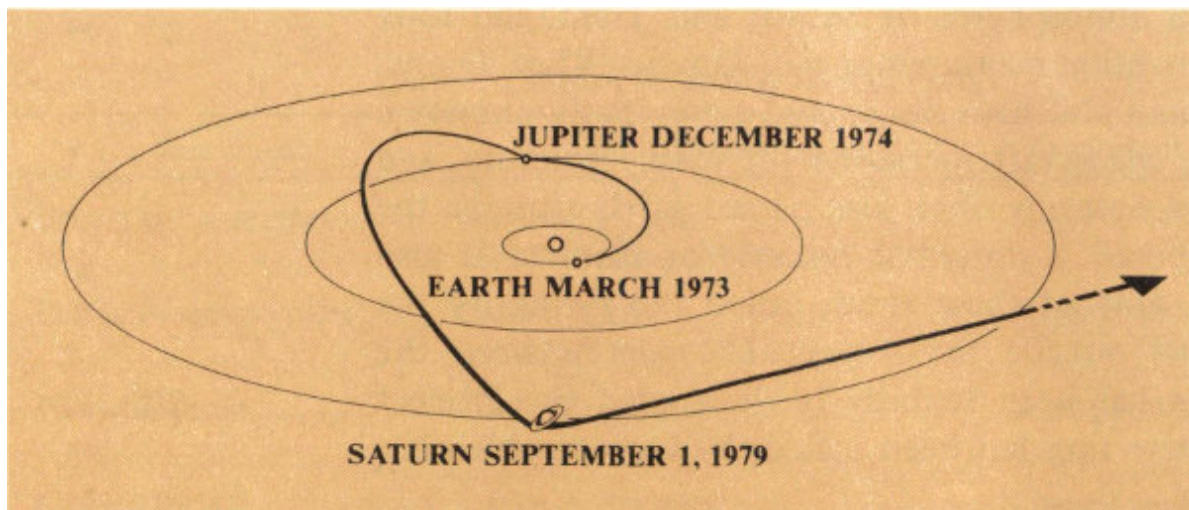
PIONEER SATURN



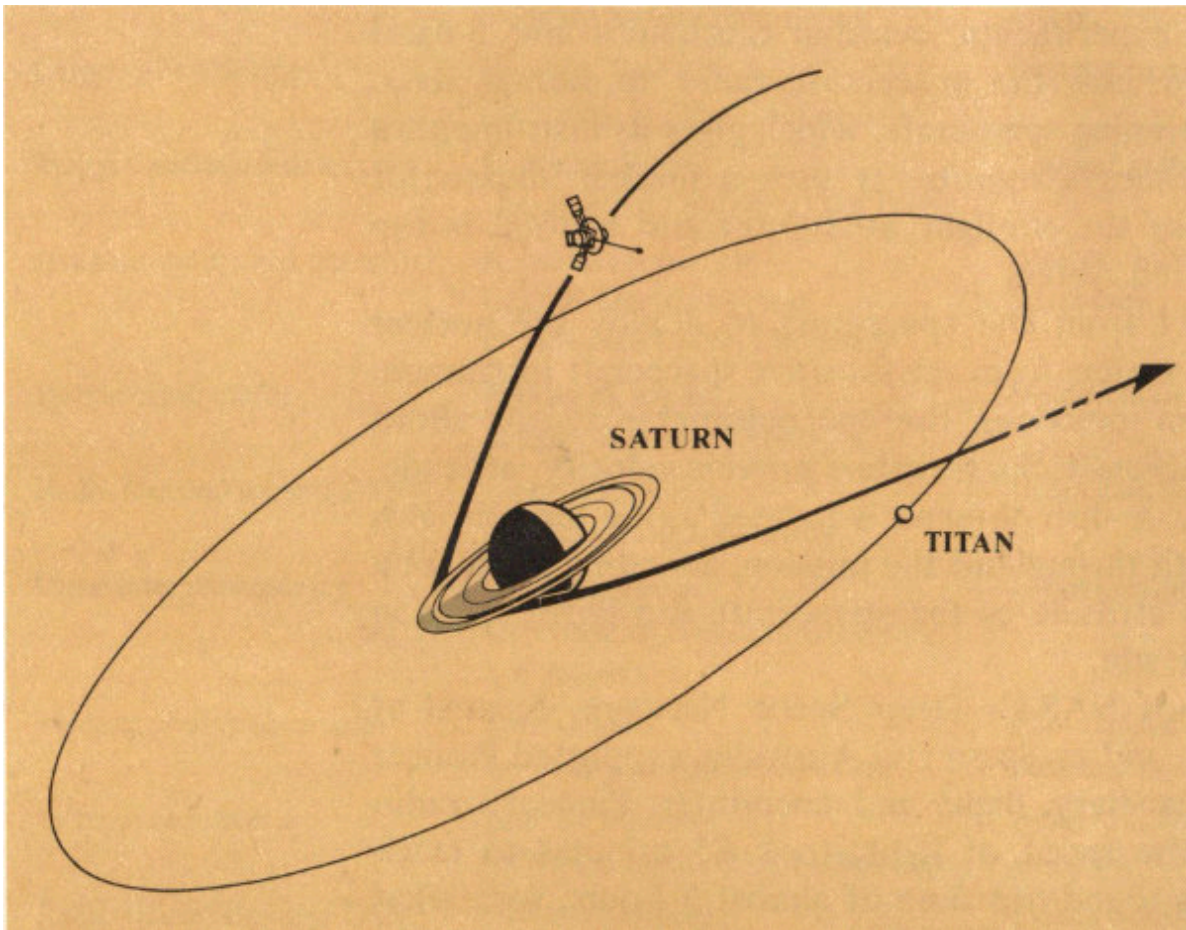
Pioneer Saturn spacecraft.

Pioneer Saturn has given us our first close view of the spectacular ringed planet Saturn and its system of moons. The spacecraft began its journey to the giant planets Jupiter and Saturn on April 5, 1973, as Pioneer 11. It reached Jupiter on December 2, 1974, passing within 42,760 km of the Jovian cloud tops and taking the only existing pictures of Jupiter's polar regions. Jupiter's massive gravitational field was used to swing Pioneer 11 back across the solar system toward Saturn. Additional maneuvers were executed in 1975 and 1976 to place the spacecraft on a suitable trajectory, with the final aimpoint selected in 1977.

From the many possible targeting options for the first Saturn flyby, two aimpoints were considered, both of which would result in a near-equatorial flyby that would give the best mapping of the high-energy particles and the magnetic field near the planet. The difference between these two aimpoints, which came to be known as the “inside” and “outside” options, was their relationship to Saturn’s unique ring system first discovered by Galileo in 1610. The “outside” option was finally selected because it was considered to be of less risk to the spacecraft and more valuable in planning the subsequent encounter of Saturn by Voyager 2, which will reach Saturn in 1981. Final targeting was completed during early 1978, when a series of timed rocket thrusts locked Pioneer into the desired trajectory. 5



Pioneer Saturn voyage.



Encounter trajectory.

On September 1, 1979, the spacecraft, now designated Pioneer Saturn, reached Saturn after 6 years in flight. It passed through the ring plane outside the edge of Saturn's A-ring and then swung in under the rings from 2,000 to 10,000 km below them. At the point of closest approach, it attained a speed of 114,100 km/h (71,900 mi/h) and came within 21,400 km of the planet's cloud tops. While it was approaching, encountering, and leaving Saturn, the spacecraft took the first closeup pictures of the planet, showing 20 to 30 6 times more detail than the best pictures taken from Earth, and made the first close measurements of its rings and several of its moons, including the largest moon, the planet-sized Titan. Titan, along with Mars, has been considered by many scientists to be the most likely place to find life in the solar system.

Pioneer Saturn unraveled many mysteries. It determined that Saturn has a magnetic field and trapped radiation belts, measured the mass of Saturn and

some of its moons, and studied the character of Saturn's interior. It confirmed the presence and determined the magnitude of an internal heat source for Saturn. Its instruments studied the temperature distribution, composition, and other properties of the clouds and atmospheres of Saturn and Titan, and took photometric and polarization measurements of Iapetus, Rhea, Dione, and Tethys. Pioneer may also have discovered a previously unknown moon of Saturn. The spacecraft measured the mass, structure, and other characteristics of Saturn's rings, and passed safely through the outer E-ring, which posed a potential hazard for Pioneer. It also discovered new rings. One of these rings, called the F-ring by the Pioneer team, lies just outside the A-ring. The gap between the F-ring and the A-ring has been tentatively designated the Pioneer Division. The other new ring has been called the G-ring, which lies well outside the F-ring.

Pioneer carries a scientific payload of 11 operating instruments; another instrument, the asteroid/meteoroid detector, was turned off in 1975. Two other experiments, celestial mechanics and S-band occultation of Saturn, use the spacecraft radio to obtain data. Pioneer Saturn is a spinning spacecraft, which gives its instruments a full-circle scan 7.8 times a minute. It uses a nuclear source for electric power because the sunlight at Jupiter and beyond is too weak for a solar-powered system.

Two booms project from the spacecraft to deploy the nuclear power source about 3 meters from the sensitive spacecraft instrumentation. A third boom positions the magnetometer sensor about 6 meters from the spacecraft. Six thrusters provide velocity, attitude, and spin-rate control. A dish antenna is located along the spin axis and looks back at Earth throughout the mission, adjusting its view by changes in spacecraft attitude as the spacecraft and Earth move in their orbits around the sun.

Tracking facilities of NASA's Deep Space Network, located at Goldstone, California, and in Spain and Australia, supported Pioneer Saturn during interplanetary flight and encounter. Pioneer's radio signals, traveling at the speed of light, took 85 minutes to reach Earth from Saturn, a round-trip time of almost 3 hours, somewhat complicating ground control of the spacecraft. Almost 10,000 commands were sent to the spacecraft in the 2-week period before closest approach. Continued communications should be possible through at least the mid 1980's. 7

After the spacecraft passed Saturn, it headed out of the solar system, traveling in the direction the solar system moves with respect to the local stars in our galaxy and in approximately an opposite direction from its sister spacecraft, Pioneer 10. Both spacecraft have plaques attached to them which contain a message from Earth for any intelligent species that may intercept the spacecraft during their endless journeys through interstellar space.

Pioneer Saturn Scientific Instruments

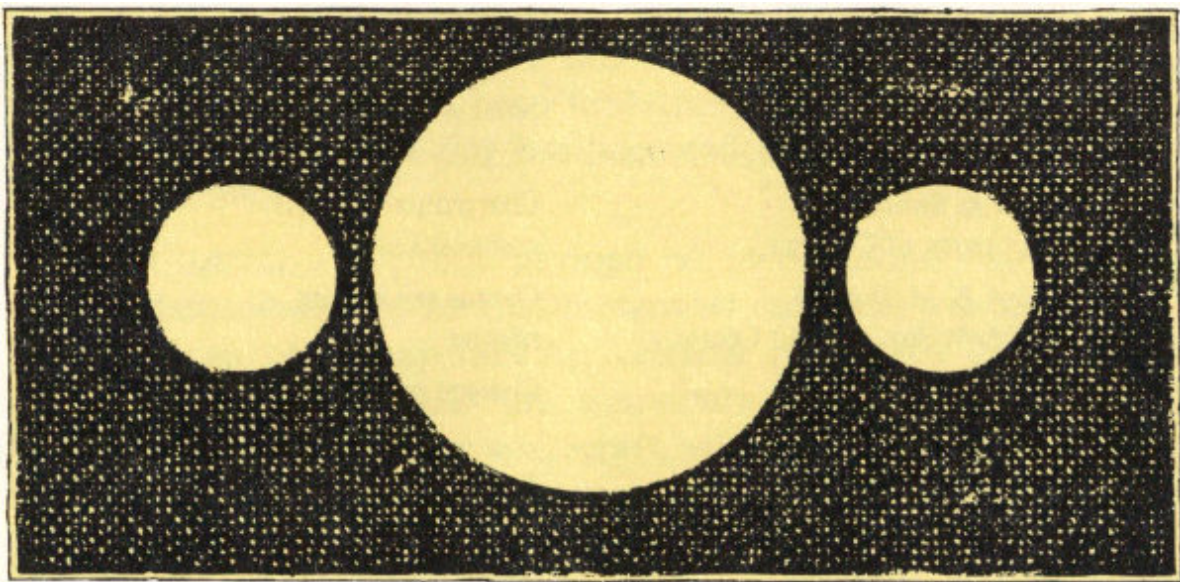
Instrument	Principal Investigator	Experiment Objective
Helium vector magnetometer	Edward J. Smith Jet Propulsion Laboratory	Magnetic fields
Fluxgate magnetometer	Mario Acuña Goddard Space Flight Center	Magnetic fields
Plasma analyzer	John H. Wolfe Ames Research Center	Solar plasma
Charged particle	John A. Simpson University of Chicago	Charged particle composition
Cosmic ray telescope	Frank B. McDonald Goddard Space Flight Center	Cosmic ray energy spectra
Geiger tube telescope	James A. Van Allen University of Iowa	Charged particles
Trapped radiation detector	R. Walker Fillius University of California, San Diego	Trapped radiation
Asteroid/meteoroid detector ^[1]	Robert K. Soberman General Electric Co. and Philadelphia Drexel University	Asteroid/meteoroid astronomy
Meteoroid detector	William H. Kinard Langley Research Center	Meteoroid detection
Radio transmitter and DSN	John D. Anderson Jet Propulsion Laboratory	Celestial mechanics

Ultraviolet photometer	Darrell L. Judge University of Southern California Los Angeles	Ultraviolet photometry
Imaging photopolarimeter	Tom Gehrels University of Arizona, Tucson	Photo imaging and polarimetry
Infrared radiometer	Andrew Ingersoll California Institute of Technology	Infrared thermal structure
Radio transmitter and DSN	Arvydas J. Kliore Jet Propulsion Laboratory	S-band occultation

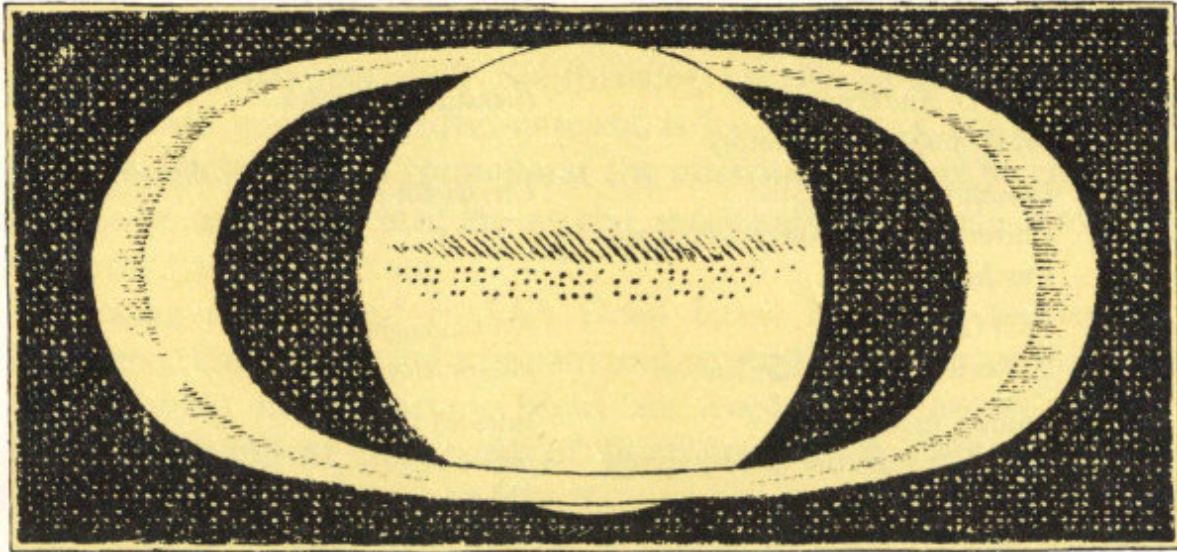
[\[1\]](#) Not currently operational.

SATURN IMAGES

Saturn, called the wonder of the heavens by early astronomers, has been studied from Earth for many centuries. When Galileo first focused his telescope on Saturn in 1610, he realized that the appearance of the planet was unusual, but he never knew its real character because the power of his homemade telescope was far too low. He thought he was looking at three globes, one large and two small, which seemed to change slowly in appearance. In 1655, Huygens, after years of observing the planet, finally realized that these projections were actually a flat ring slightly separated from the main globe.

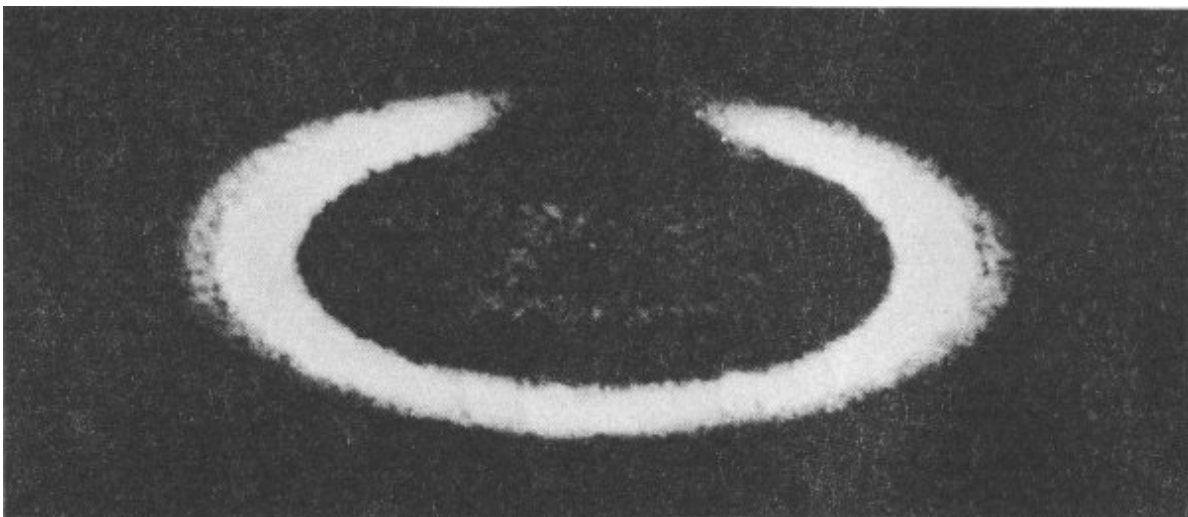


First sketch of Saturn (Galileo, 1610).



First sketch showing a division in the rings (Cassini, 1675).

In 1675, Cassini found the first breach in the supposedly solid, rigid, and opaque ring when he discovered that it was divided into two parts by a dark line, now known as Cassini's Division. In later years he also detected some of Saturn's moons. 9



First successful photograph of Saturn (Andrew Common, 1883). (Discovering the Universe, Colin A. Ronan, Copyright 1971 by Colin A. Ronan, Basic Books, Inc., Publishers, New York.)

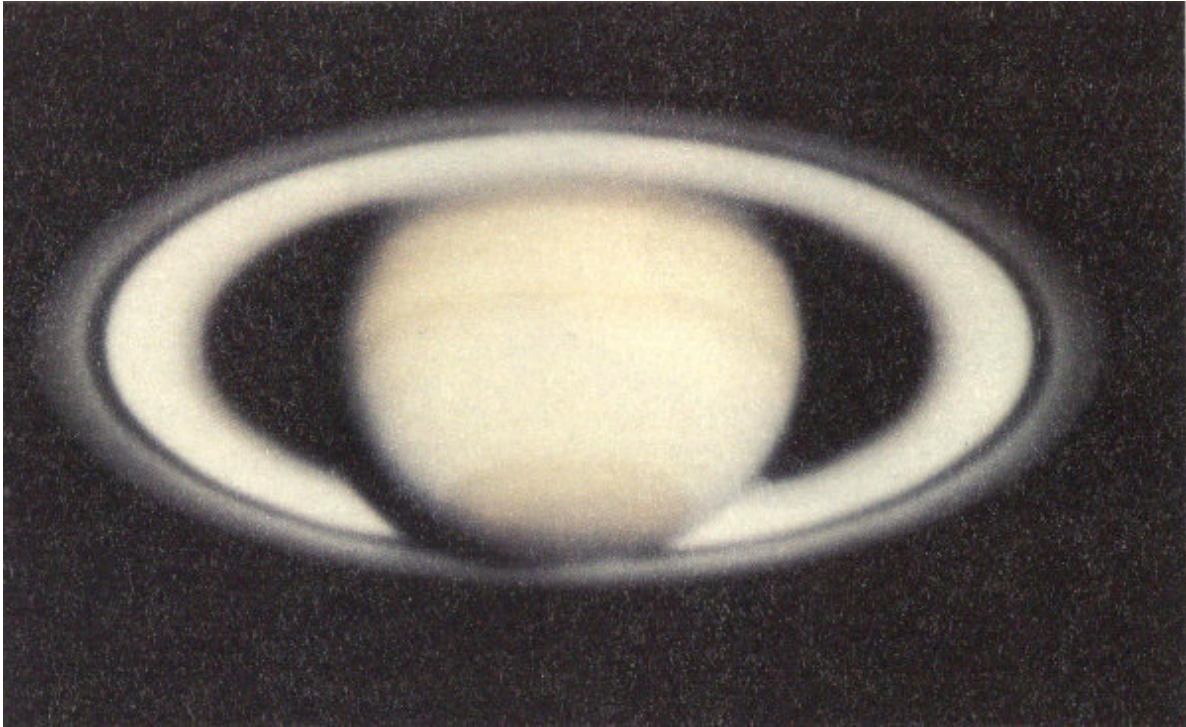
The earliest successful photograph of Saturn was taken in 1883 by Andrew Common. In 1895, James Keeler suggested that the rings are in fact a swarm

of particles in near-independent orbits. These rings, until the recent discoveries of faint ring systems around Jupiter and Uranus, were considered unique in the solar system.

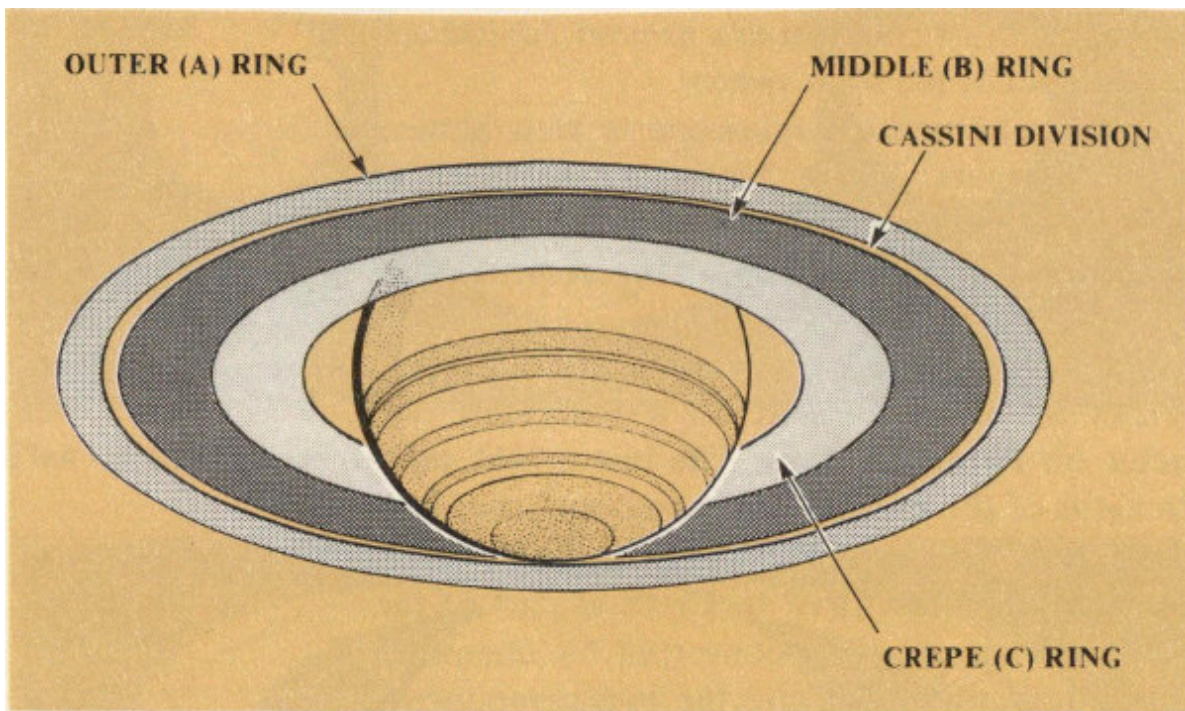
Since Galileo first used his homemade telescope to view Saturn, there have been many observers. There have also been major advances in telescopes; a resulting modern view of Saturn is shown on the next page. The corresponding sketch shows the nomenclature of the brightest rings.

Now the planet has been seen for the first time not from Earth, but in much closer views by an instrument on a spacecraft, the imaging photopolarimeter on Pioneer Saturn. The instrument separately measures the strengths of the red and blue components of sunlight scattered from the clouds of Saturn and converts this information into numbers. The data are transmitted to Earth as part of the spacecraft telemetry. The signals are then converted by computer into shades of gray on photographic film, and the two components plus a synthesized green image can be recombined into a color image that approximates the planet's true color. Some of the resulting images are shown on the pages following the Earth-based view. These pictures were produced by a scientific team from the University of Arizona.

These images, while helping to unravel some of the mystery surrounding the planet, have created even more interest regarding it. We are really just beginning to know Saturn. It is up to future spacecraft to more completely reveal her secrets and solve her mysteries.



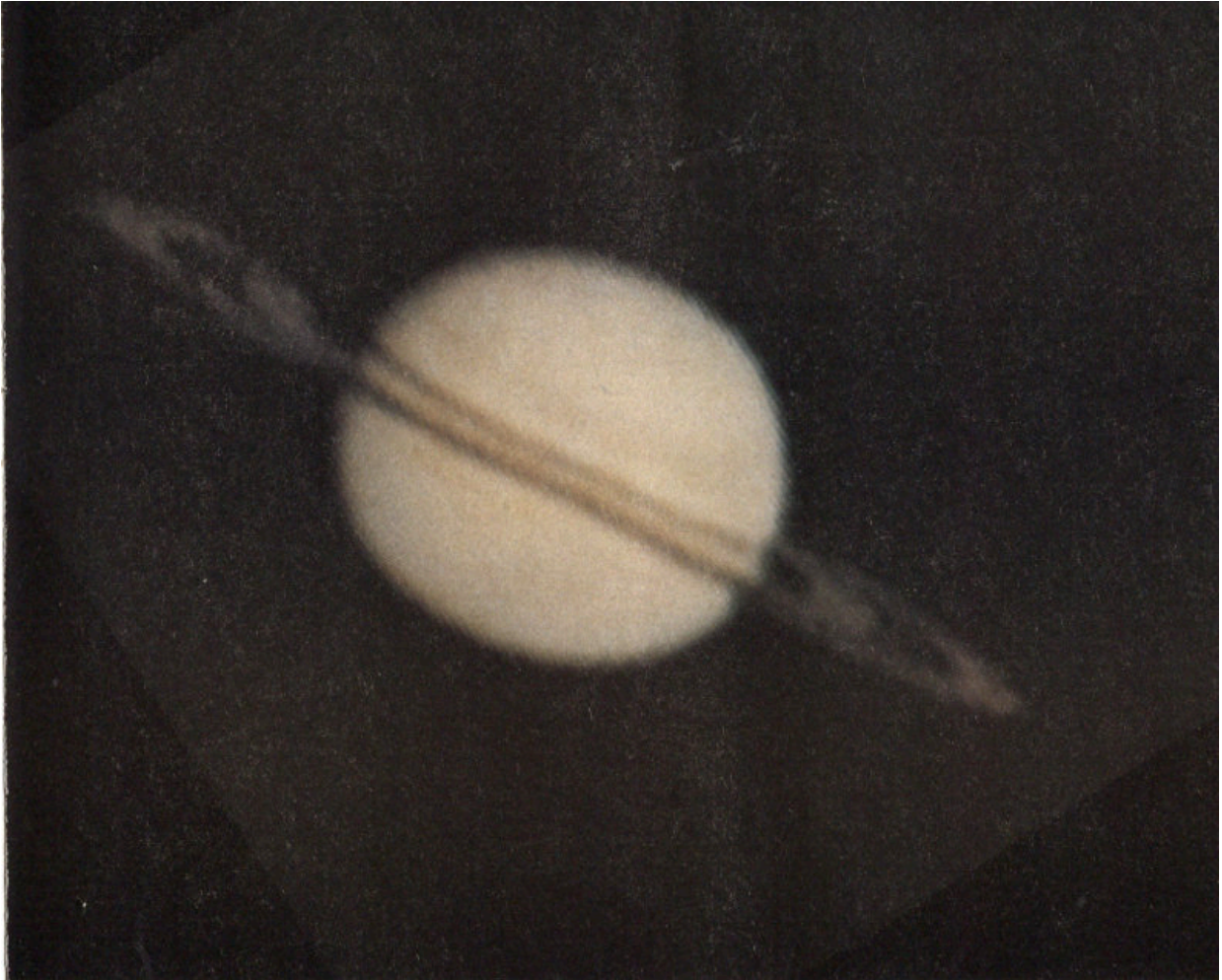
High-quality contemporary Earth-based view of Saturn (Photo: Catalina Observatory, University of Arizona).



Nomenclature of bright rings.

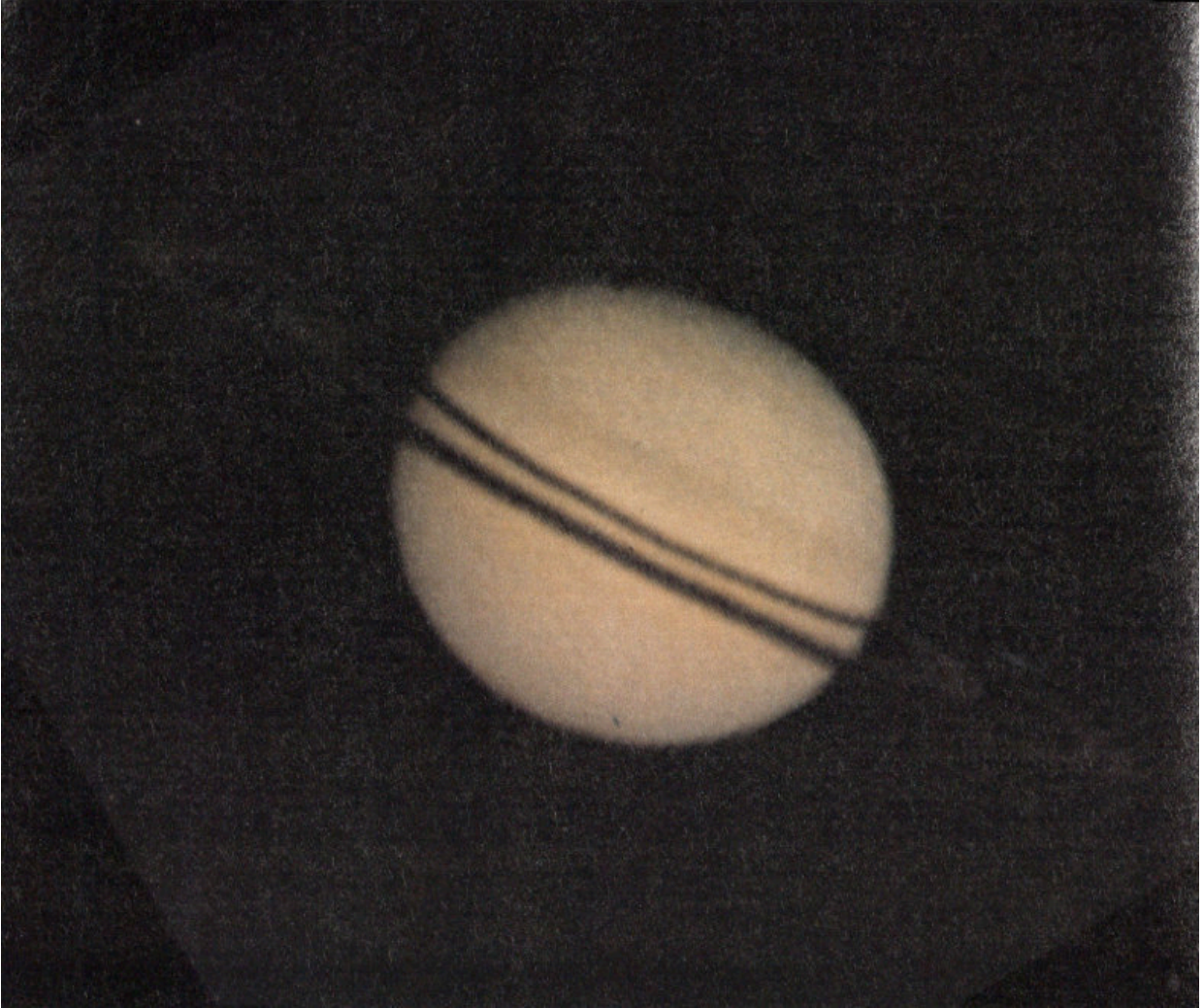
OUTER (A) RING
CASSINI DIVISION
MIDDLE (B) RING
CREPE (C) RING

11

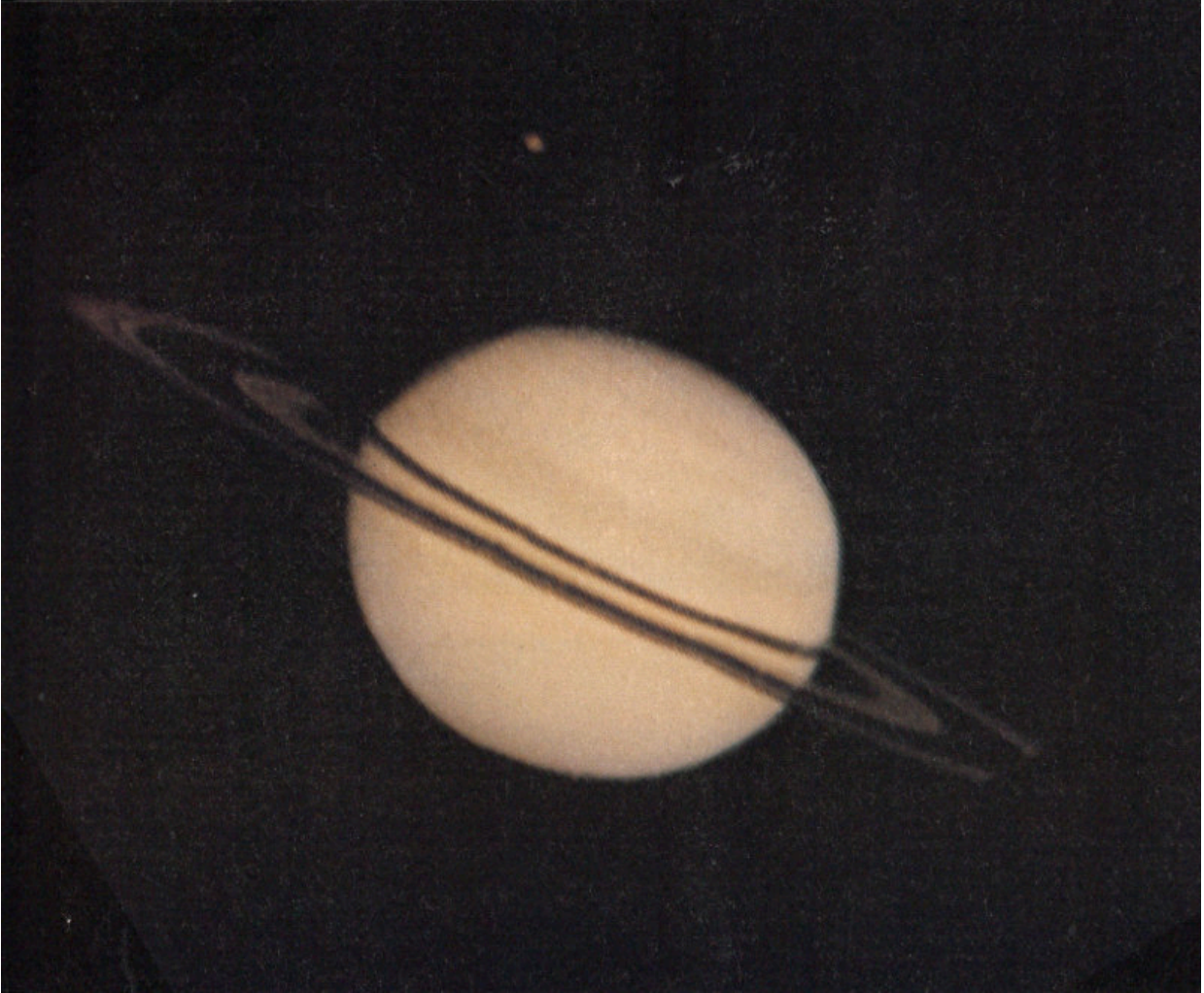


Pioneer Saturn image of Saturn and its rings from a distance of 8,400,000 km (August 22, 1979, 10 days before closest approach). Resolution has not yet reached Earth-based quality. Faint banding is visible on the disk of the planet. The silhouette of the rings can be seen in front of the planet. Slightly above this silhouette is the shadow of the rings on the disk. Beyond the disk, structure can be seen in the rings. The rings have a distinctly different appearance in this and subsequent images than in Earth-based pictures because they are illuminated from below rather than from above as we view them from Earth. The A-ring and C-ring are bright, and between them the B-ring is dark. The Cassini Division at the inner edge of the A-ring is also bright, but is blended with the ring at this distance.

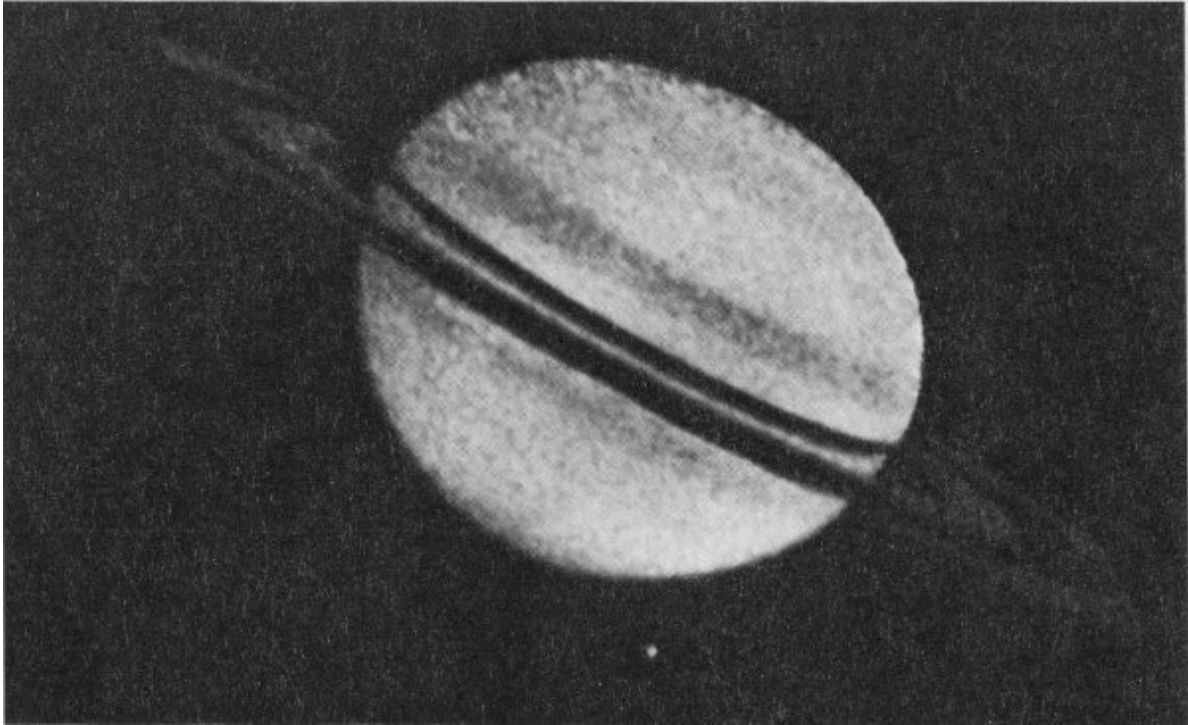
12



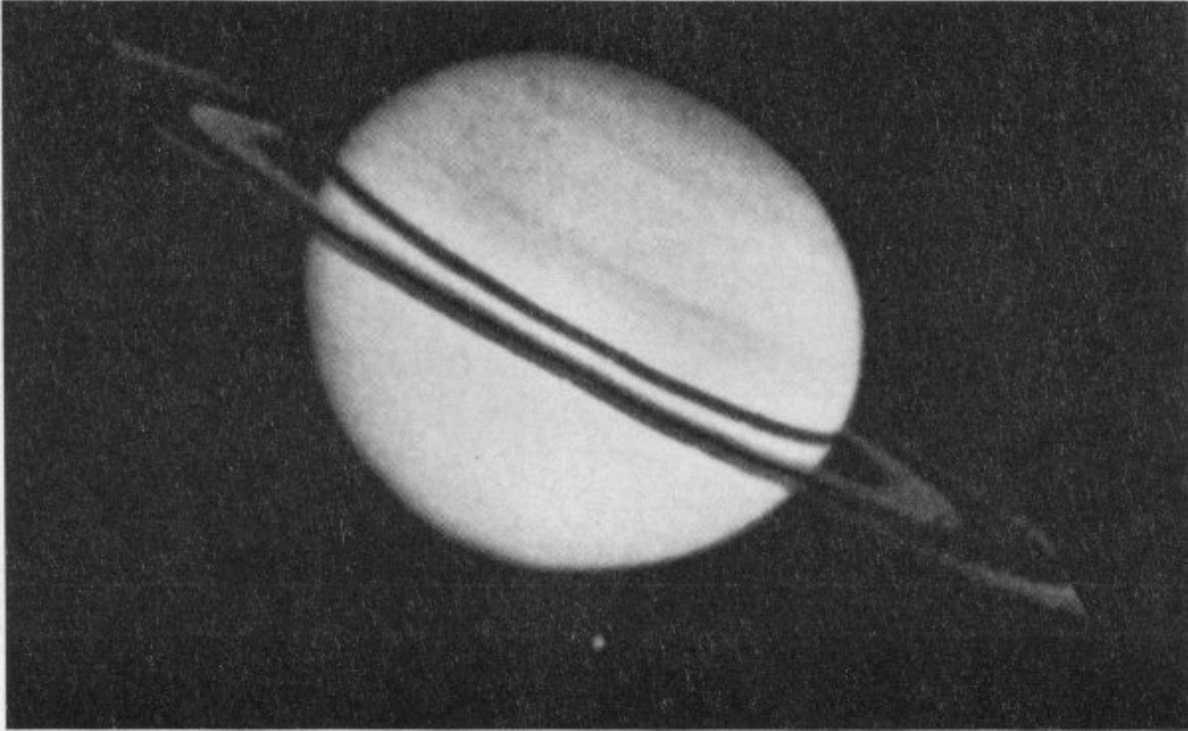
Pioneer Saturn image of Saturn and its rings from a distance of 5,500,000 km (August 26, 1979, 6 days before closest approach). Resolution of features is beginning to be approximately equal to that of Earth-based pictures. Belts on the planet are becoming more distinct, and considerable structure can be seen in the rings. The small blue spot at the bottom of the planet is due to incomplete data and will be corrected by further processing. A notch appears at the top left of the planet, which may be a data transmission problem or the shadow of one of Saturn's moons.



Pioneer Saturn image of Saturn and its rings from a distance of 2,800,000 km (August 29, 1979, 72 hours before closest approach). The Cassini Division at the inner edge of the A-ring is clearly resolved and is bright. The A-ring is dark outside the Cassini Division because it has more particulate matter there. Polar belts are becoming more visible on the face of the planet. Irregularities that appear in the ring silhouette and shadow are due to stepping anomalies in the imaging photopolarimeter and will be removed in further processing. The small round image that appears above the planet is the moon Titan.

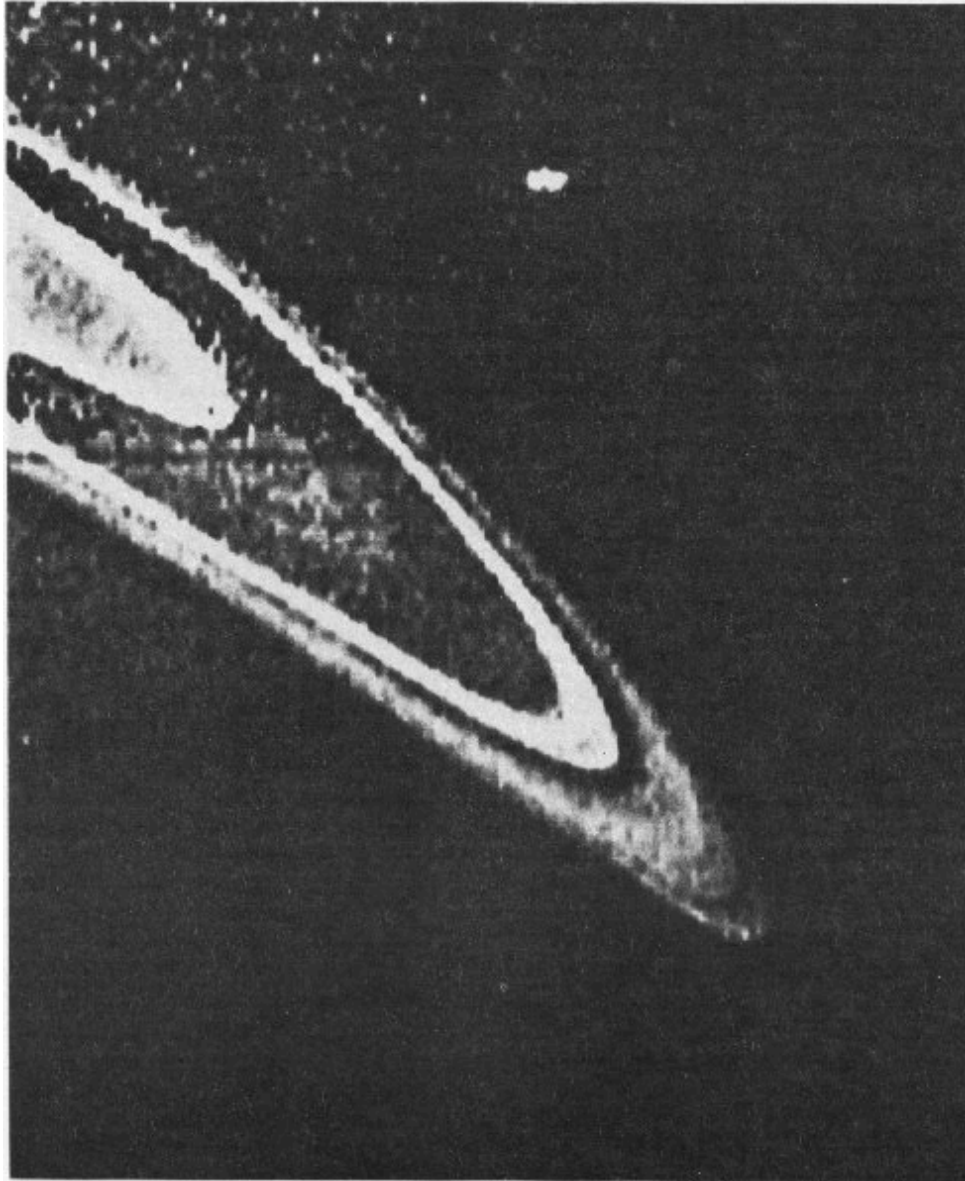


These pictures of Saturn and its rings were taken by Pioneer Saturn from a distance of 2,500,000 km (August 29, 1979, 58 hours before closest approach). The imaging photopolarimeter gathers data using the red and blue components of the light reflected from Saturn. These two views are from the two color components (upper, blue; lower, red). The banded structure on the planet is particularly evident in the upper image. Because the spacecraft is nearing the planet, the rings are partially outside the field of view of the instrument. The speck of light below the planet is Saturn's moon Rhea, which is 1450 km in diameter, about one-half the size of Earth's moon.

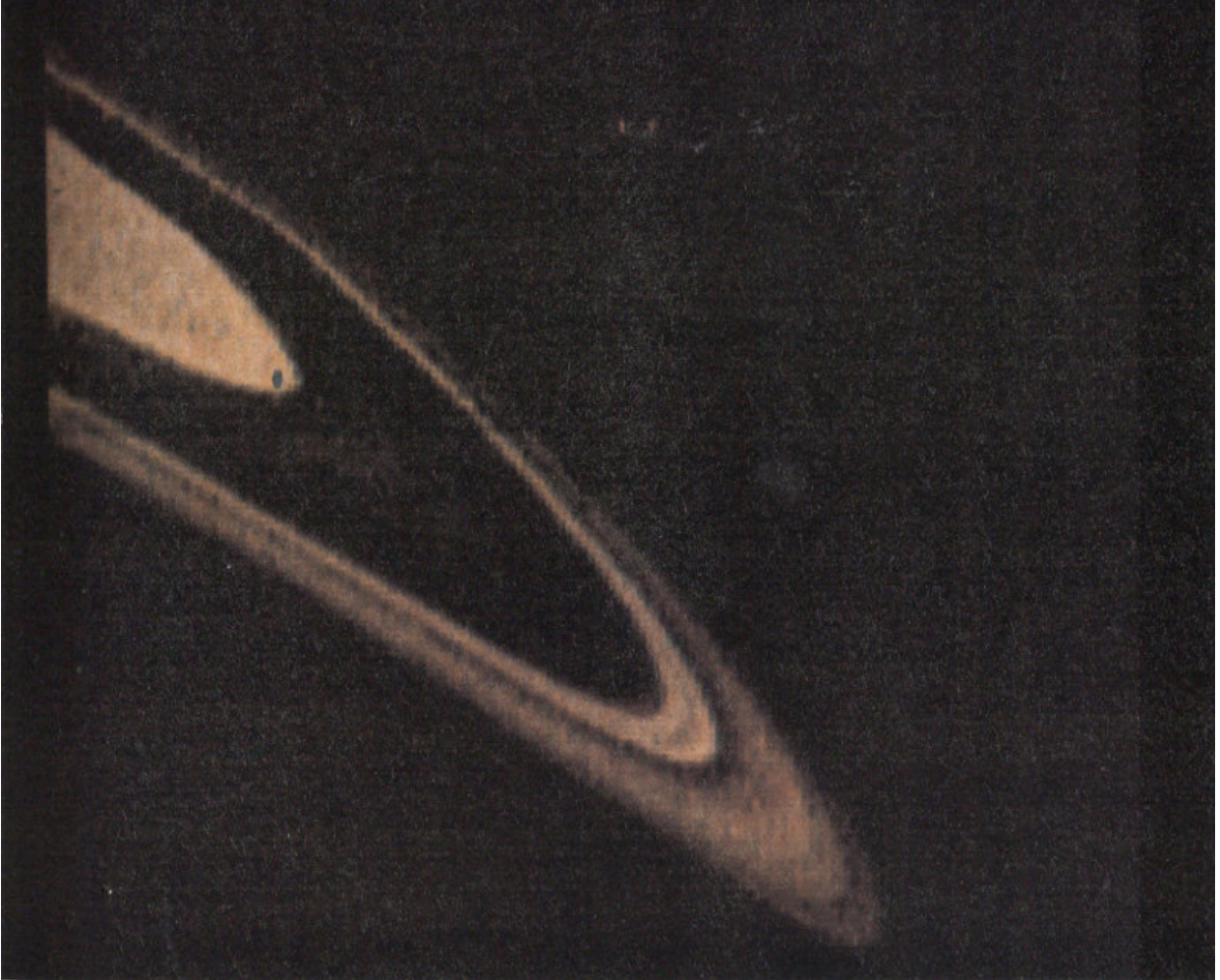




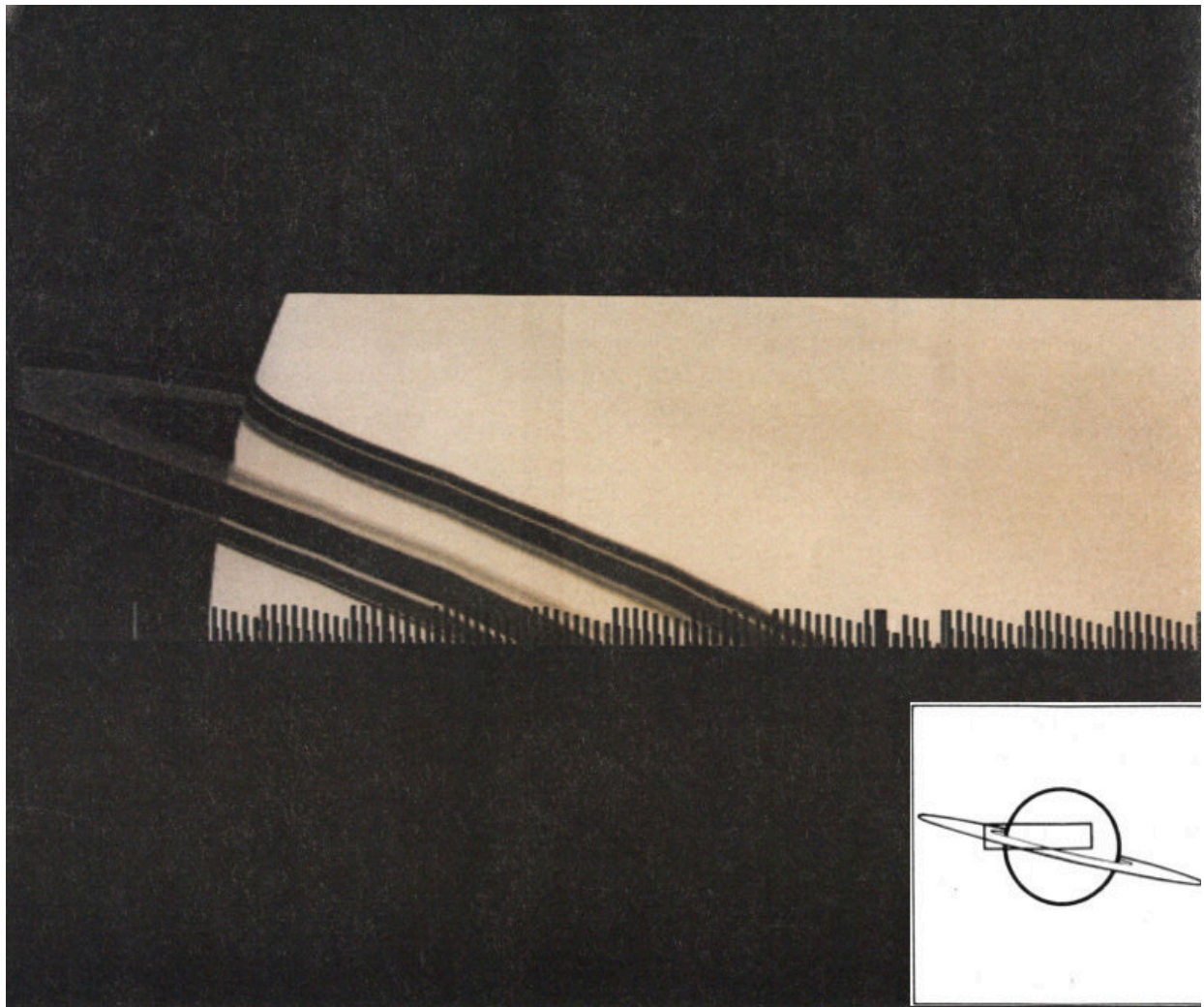
This image was produced by combining the two images on the facing page, adding a synthesized green component, and adjusting the intensity of each to obtain an approximation to Saturn's true color. (The same process was used in all of the Pioneer color images of Saturn shown here.) Although not evident on this reproduction, scientists believe that, from detailed study of both the uncombined and the combined images, they can begin to see evidence of jet streams in Saturn's upper atmosphere. The slight blue edge is an artifact from the computer-enhancement procedure.



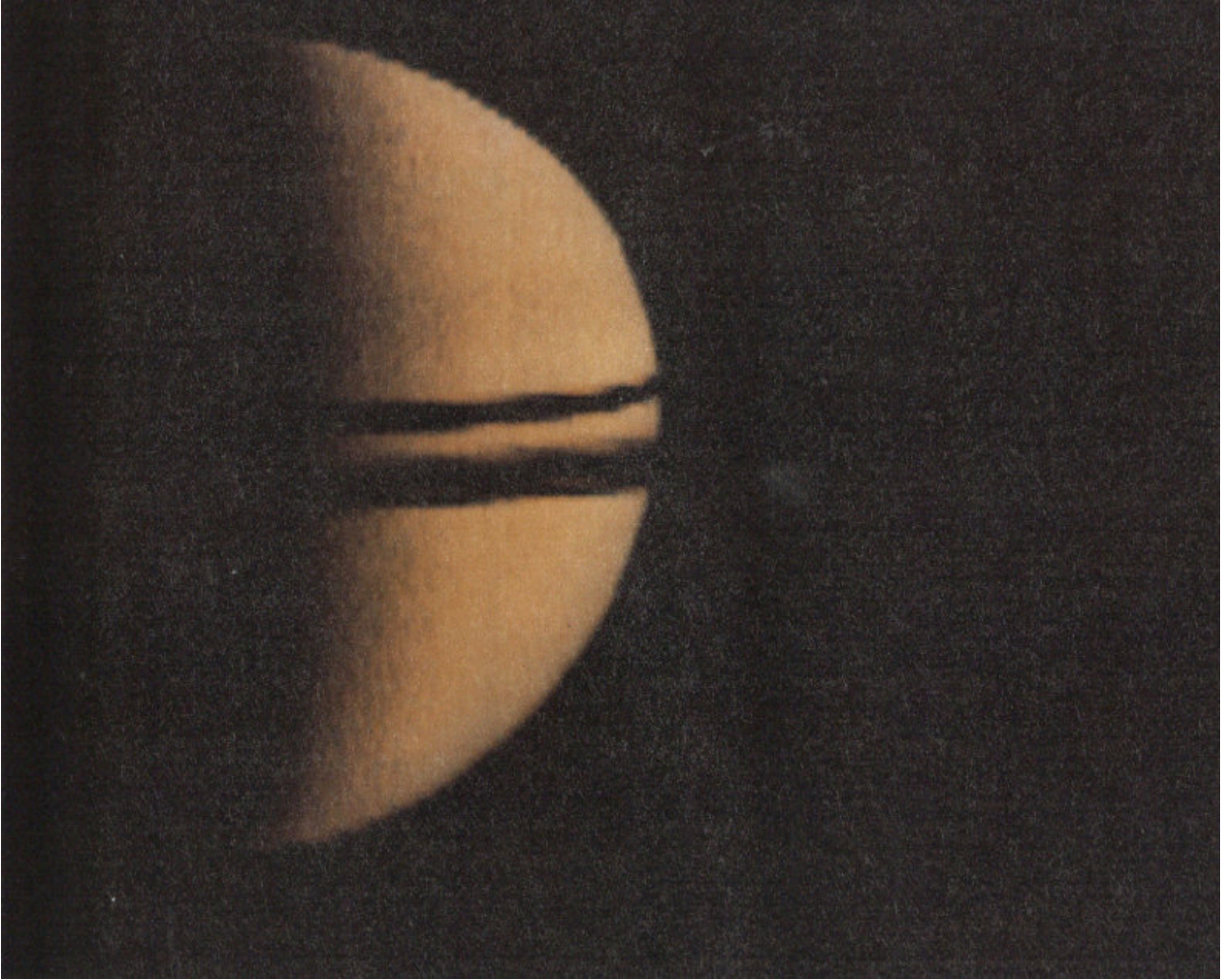
Pioneer Saturn view showing the structure of Saturn's ring system in detail never before seen. The image was taken from a distance of 943,000 km (August 31, 1979, about 44 hours before encounter). The moon Tethys, seen at the top of the image, is 1200 km in diameter. There is a very faint unidentified Saturnian moon at the lower right, just off the tip of the bright A-ring (may not be visible in this print). The newly defined F-ring appears faintly just outside the bright edge of the A-ring. The region between the A-ring and the F-ring has been tentatively named the Pioneer Division. This print has been processed to enhance detail of the main rings.



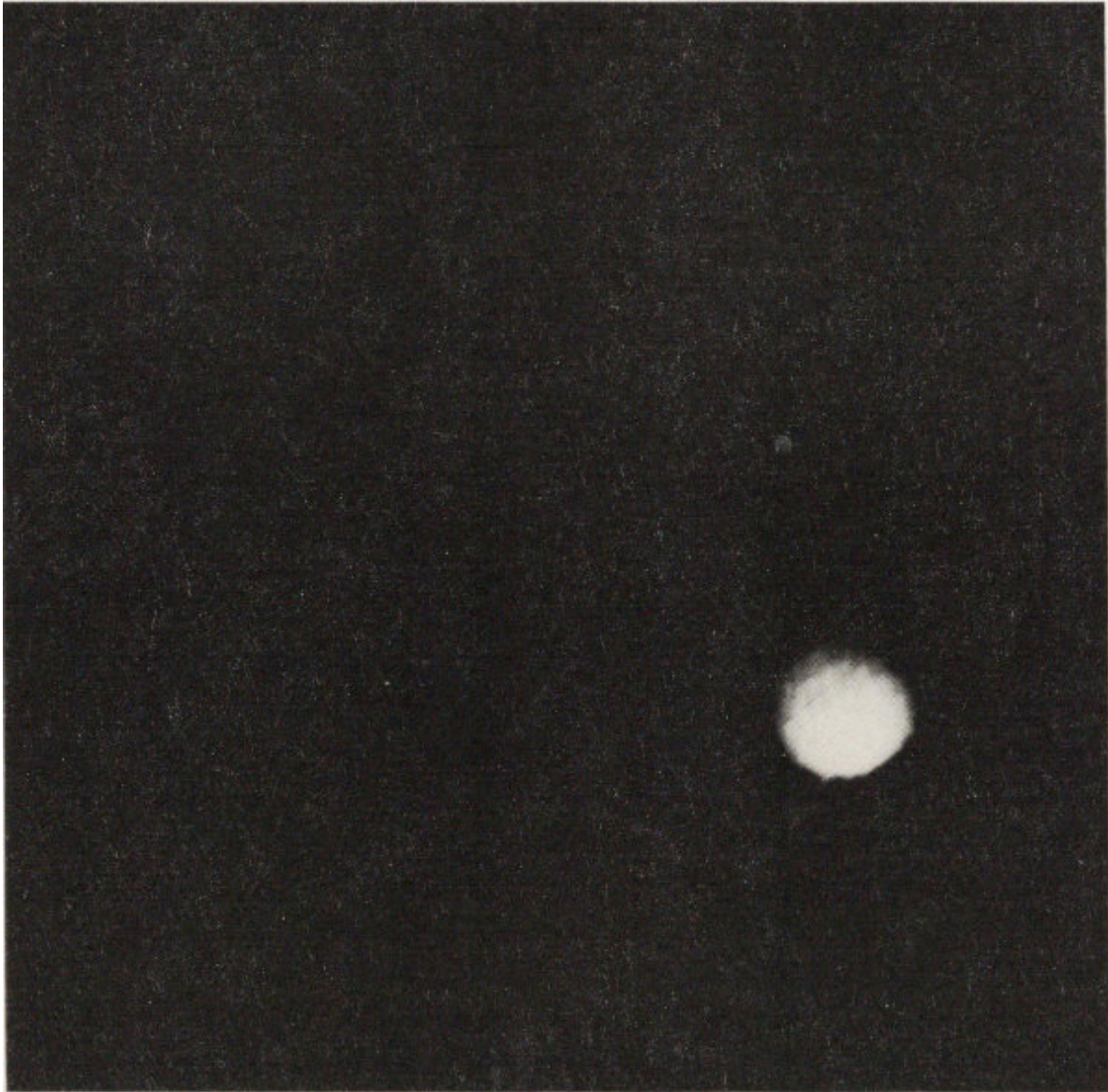
A color print of a version of the same image as shown on the facing page. The blue dot at the outer edge of the C-ring is an artifact. This image has not been computer processed to the same extent as the facing image. Tethys, for example, is only faintly visible, and the F-ring cannot be seen.



Close-up image from Pioneer Saturn of Saturn from a distance of 400,000 km (September 1, 1979, a few hours before closest approach). Inset shows the location of the image on the planet. The part of the disk shown is about 25,000 by 70,000 km. The sawtooth pattern is an artifact. The vertical stripe on the disk is due to a gain change in the instrument. The image has not had its final corrections for shape. The rings and their shadows cut diagonal swaths across the image with the upper swath being the shadow. The rings, seen from the unlit side, are visible in the foreground. The ring shadow shows clearly that Saturn's rings have two major divisions: the Cassini Division dividing the outer A-ring from the middle B-ring, and a second division (previously controversial) dividing the B-ring from the inner C-ring. These divisions show as parallel pinstripes in the broad black band of the rings' shadow, with the upper stripe being the Cassini Division. Some shearing in the bands and belts on Saturn's disk is beginning to appear, although the low contrast on the planet (due to its high-altitude haze) does not make this highly evident.



Post-encounter image of Saturn from a distance of 850,000 km (September 2, 1979, 15 hours after encounter). As planned, during the encounter Saturn's gravitational field turned the spacecraft's trajectory behind the planet and out toward the edge of the solar system at approximately a right angle to the inbound trajectory. Thus, the planet is now illuminated from the side, and the view is of a crescent Saturn with the terminator on the right of the picture. The rings appear dim when compared with the previous inbound pictures because of the different angle between the sun, the rings, and the spacecraft. When this picture was taken, the spacecraft, as seen from Earth, was about to pass behind the sun, and solar activity was interfering with spacecraft communications. The image quality, therefore, is not as good as for pictures taken inbound. With this farewell image from Pioneer, Saturn waits for the Voyager spacecraft in 1980 and 1981.



This image of cloud-covered Titan is one of the “firsts” for the Pioneer Saturn mission. Titan is the largest of Saturn’s moons. Because of its great distance from Earth, however, Titan can be seen only as a point of light in Earth-based pictures. The Pioneer trajectory could not be adjusted to allow passage close to Titan, and imaging of the planet from the spacecraft was at the outer range of capability of the imaging instrument. Thus, the fuzzy edges and contrast variations on the moon should not be construed as surface features.

Titan was imaged in the spirit of exploration and discovery. With this image, which is now a part of recorded history, our visual information on Titan is materially increased, but is still only roughly equal to Galileo’s information on Saturn itself in 1610 at the time that he prepared his first sketch. Better images of Titan will be obtained by the Voyager spacecraft.

SCIENCE HIGHLIGHTS

Pioneer Saturn has already greatly expanded our knowledge of Saturn, its rings and moons. We now know that Saturn, in many ways, represents an intermediate case between Jupiter, the largest planet in the solar system, and Earth. The composition of Saturn's interior is essentially the same as Jupiter's, differing only in the size and extent of the various internal layers. Measurements for Saturn are consistent with a central core of molten heavy elements (probably mostly iron) which is the approximate size of the entire Earth, but about three times more massive. Surrounding the central core is an outer core of highly compressed hot, liquefied volatiles such as methane, ammonia, and water. This outer core is equivalent to approximately nine Earth masses. These core regions, however, represent a very small fraction of the planet, which is composed primarily of the very lightest gases, hydrogen and helium, and is almost 100 times the mass of the Earth. Because of the high pressure in Saturn's interior, the hydrogen is transformed to its liquid metallic state. Above this metallic hydrogen shell are liquid molecular hydrogen and Saturn's gaseous atmosphere and clouds, which make up the rest of the planet.

Electrical currents set up within the metallic hydrogen shell produce Saturn's magnetic field, which was measured by Pioneer. In spite of Saturn's large size, the magnetic field at the cloud tops is only slightly weaker than the field at the Earth's surface. Saturn is unique in that its magnetic axis is nearly aligned with its rotation axis, unlike Earth and Jupiter.

Saturn's magnetic field is also much more regular in shape than the fields of the other planets. At large distances from Saturn, the magnetic field is deformed by the inward pressure of the solar wind. Near the noon meridian (close to the inbound Pioneer trajectory), the solar wind causes a compression of the field; in the dawn meridian (close to the outbound trajectory), the field is swept back and presumably forms a long magnetic tail. In both cases, Pioneer Saturn crossed the outer boundary of the magnetic field several times as the field moved in and out, responding to

changing solar wind pressure. Pioneer also observed inward and outward boundaries.

The magnetic envelope surrounding Saturn is intermediate in size and energetic particle population between those of the Earth and Jupiter, the only two other planets known to be strongly magnetized. The three other planets investigated thus far (Venus, Mars, and Mercury) and Earth's moon have little or no magnetism. Virtually all our knowledge of Saturn's magnetic environment has been obtained by Pioneer. The spacecraft found rings of particulate material and several small moons near the rings, which strongly affect Saturn's trapped radiation. These features provide important diagnostic capabilities. A unique finding is the nearly total absence of radiation belt particles at distances closer to the planet than the outer edge of the visible rings. 22

The inner region of the thick magnetic envelope of Saturn, called the magnetosphere, contains trapped high-energy electrons and protons, with some evidence for heavier nuclei. The overall form of the magnetosphere is simple and compact, more similar to that of Earth than of Jupiter. The unique measuring capabilities of the Pioneer Saturn radiation detectors led directly to the discovery of a diffuse new ring of particulate matter in the region from about 10 to 15 planetary radii (1 Saturn radius = 60,000 km) from Saturn. This ring has been tentatively designated as the G-ring. The G-ring clearly causes particle absorption near the equatorial region. Moreover, Pioneer discovered another region inside 7 or 8 planetary radii in which the radiation belt particles were subjected to a strong loss or absorption, presumably caused by the presence of an extensive cloud of plasma corotating with the planet.

Inside about 10 planetary radii, the trapped radiation shows a high degree of axial symmetry around Saturn and is consistent with the centered dipole magnetic field observed by Pioneer. Saturn's rings annihilate all trapped radiation at the outer edge of the A-ring, leaving a shielded region close to the planet in which the radiation intensity is the lowest so far encountered in this mission. This shielding prevents the further buildup of electron intensities at lower altitudes, which otherwise would have been present and would have made Saturn a strong radio source observable from Earth.

Pioneer found that several of Saturn's moons absorb trapped particles from the radiation belts, producing prominent dips in the intensity. The effectiveness of absorption at the moons Tethys and Enceladus is particularly astonishing, and supports the idea that radiation belt particles are drifting inward slowly across the moons' orbits.

A precipitous decrease in particle intensity, lasting only for about 12 seconds, was observed over a wide range of energies for both protons and electrons at a distance near 2.53 Saturn radii, 23 minutes after Pioneer crossed the Saturn ring plane inbound. At about the same time, an anomaly was also observed in the magnetic field measurements. These phenomena have been tentatively interpreted as indicating the presence of a nearby massive body absorbing the trapped radiation and perturbing Saturn's magnetic field. The estimated radius of this object lies in the range of 100 to 300 km, based on the effectiveness with which it absorbed the high-23 energy radiation. The total radiation dose received at Saturn was equivalent to only 2 minutes in the Jovian radiation belts because Saturn's radiation belts were so much weaker.

In addition to images of Saturn, the brightness, color, and polarization of the reflected light were also measured by the imaging photopolarimeter on Pioneer Saturn. These measurements are used to study the cloud layers of Saturn and Titan and to model the vertical structure of the atmospheres of these two bodies. In the scans that made the images, the banded structures of Saturn and of the rings were obtained in fine detail. These are essential in studying the atmosphere, rings, and moons. A new Saturnian ring, which has been tentatively designated the F-ring, was discovered in the images. It is narrower than 500 km in width, but is important because it forms an outside barrier to the bright A- and B-rings. The gap between the F- and A-rings has been designated the Pioneer Division by the Pioneer team. A small moon, which either was previously unknown or had been previously discovered from Earth but lost again, was found in the Pioneer Saturn images. After its initial discovery, this new moon continued on its 17-hour orbit around the planet and passed near Pioneer as the spacecraft entered the ring system. It is quite conceivable that this moon is the same one that perturbed the radiation belt particles and produced the anomaly in the magnetic field measurements.

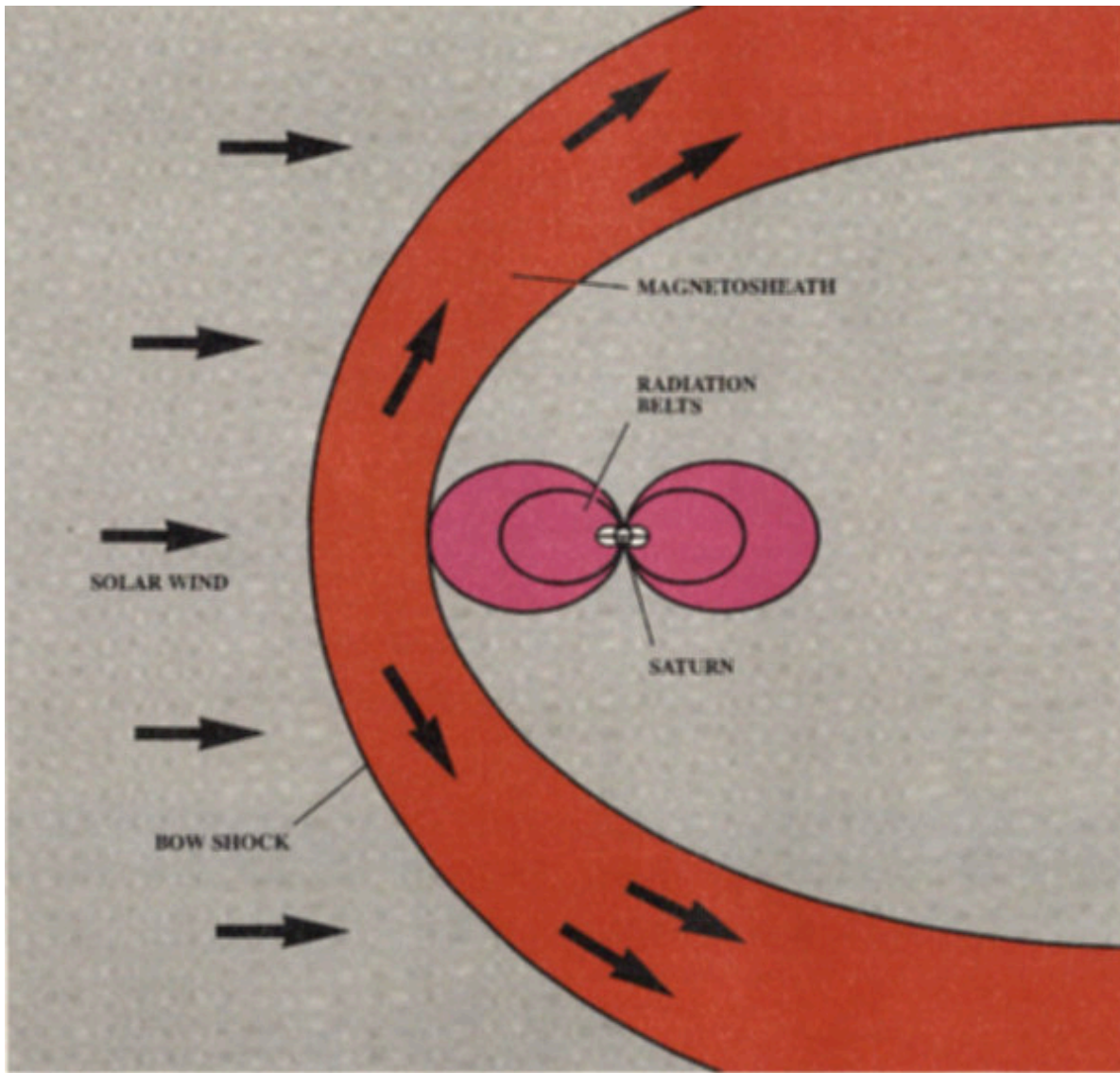
Infrared observations obtained during the Saturn flyby revealed the temperatures in the atmospheres of Saturn and the rings and in the atmosphere of Titan. It was found that Saturn has a temperature of about 100 K (about 280°F below zero) and, according to these observations, has an internal heat source of enough strength that the planet emits approximately 2.5 times as much energy as it absorbs from the sun. The equatorial yellowish band observable in many of the images was found to be several degrees colder than the planet at other latitudes and is probably a zone of high clouds resembling similar zones on Jupiter. As expected, the rings were extremely cold, 65 to 75 K (about 330°F below zero), at the time of encounter. The temperature differences between the illuminated and unilluminated sides of the rings, and the rate of cooling as the ring particles go into Saturn's shadow, suggest that the ring particles are at least several centimeters in diameter and the rings themselves are many particle diameters thick. The very minor perturbation to Pioneer's trajectory, as it passed under the visible rings, indicates that the rings probably consist of ices.

As Pioneer passed through Saturn's ring system, very sensitive meteoroid detectors observed the impact of five particles on the spacecraft, particles that were about 10 micrometers (0.0005 inch) in diameter. Two 24 impacts occurred while the spacecraft was above the rings and three while the spacecraft was below the rings. No impacts were detected going through the ring plane, but the Pioneer instrument cannot detect individual impacts that occur less than 77 minutes apart. This characteristic would have prevented detection of ring particles because of the impacts detected just before both ring plane crossings. It is uncertain whether the micrometeoroids detected by Pioneer Saturn were stray ring particles deflected out of Saturn's ring plane or whether they were particles from interplanetary space drawn inward toward Saturn by its strong gravitational field.

Close to the point of closest approach to Saturn, the spacecraft's radio transmissions were affected by Saturn's ionosphere. The manner in which the radio signals were absorbed indicates that Saturn has an extensive ionosphere composed of ionized atomic hydrogen with a temperature of about 1250 K in its upper regions. This high temperature requires an extensive energy source other than the sun. This phenomenon was also observed at Jupiter.

Pioneer measured ultraviolet glow throughout the Saturnian system. This ultraviolet glow is due to the scattering of the light from the sun by atomic hydrogen. The observations of ultraviolet emission from an extensive cloud of hydrogen gas surrounding Saturn's visible rings are especially interesting. The rings themselves are presumably the source of this hydrogen. On the planet's disk, the ultraviolet observations show significant latitude variations, suggesting the possibility of aurora near Saturn's polar regions. A similar extensive cloud of hydrogen was also seen partially surrounding Titan's orbit.

These very preliminary findings by Pioneer Saturn represent only a small fraction of what will ultimately be learned about Saturn and its environment as the spacecraft data are analyzed in greater detail over the weeks and months ahead.



Schematic of the solar wind interaction with Saturn's magnetosphere. The solar wind arrives from the direction of the sun, is deflected at Saturn's bow shock, and flows around Saturn in the magnetosheath (orange region), as indicated by the arrows. The sizes of the magnetosheath and radiation belts change in response to the external solar-wind pressure, becoming smaller when the external pressure is larger and vice versa.

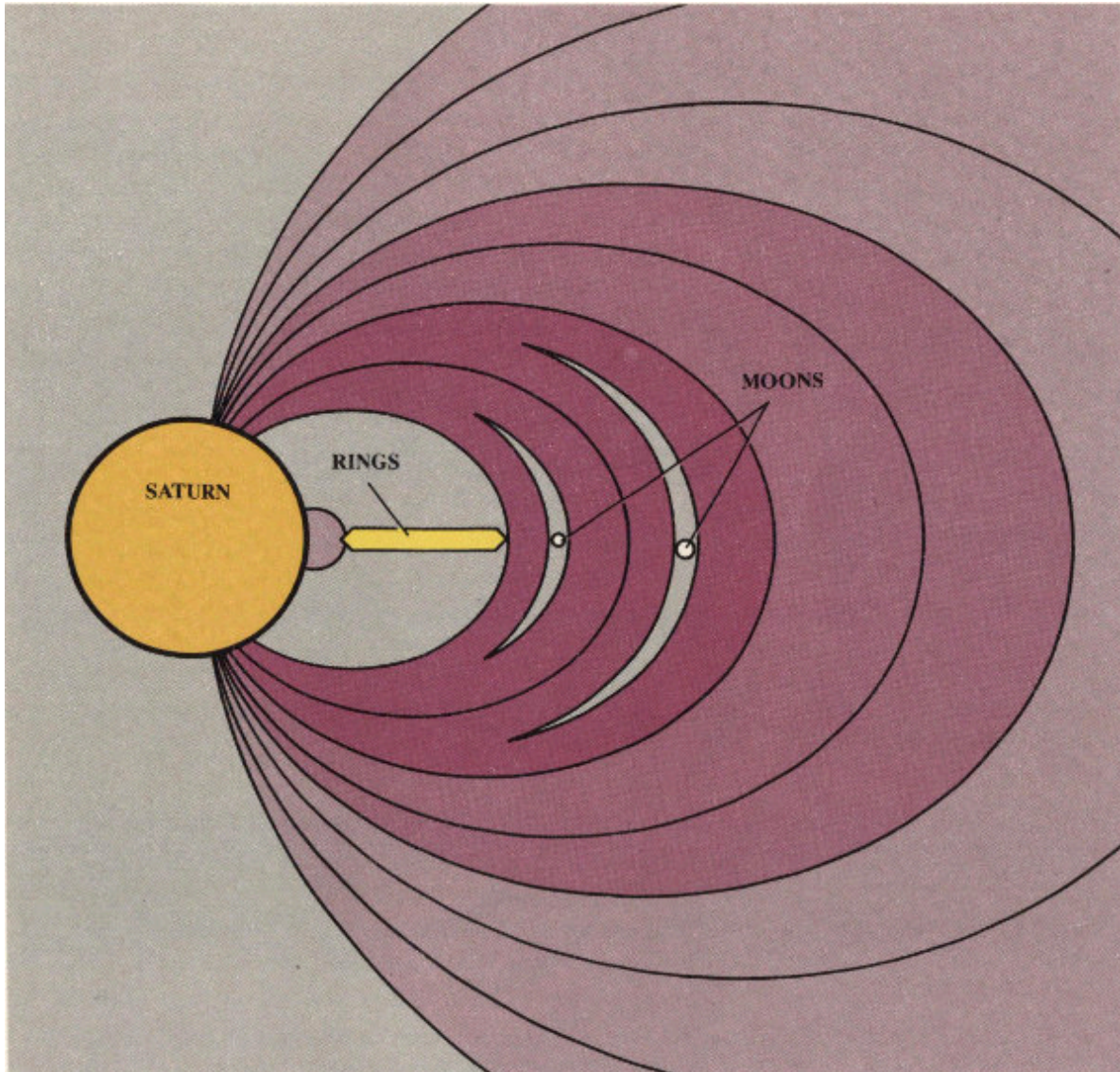
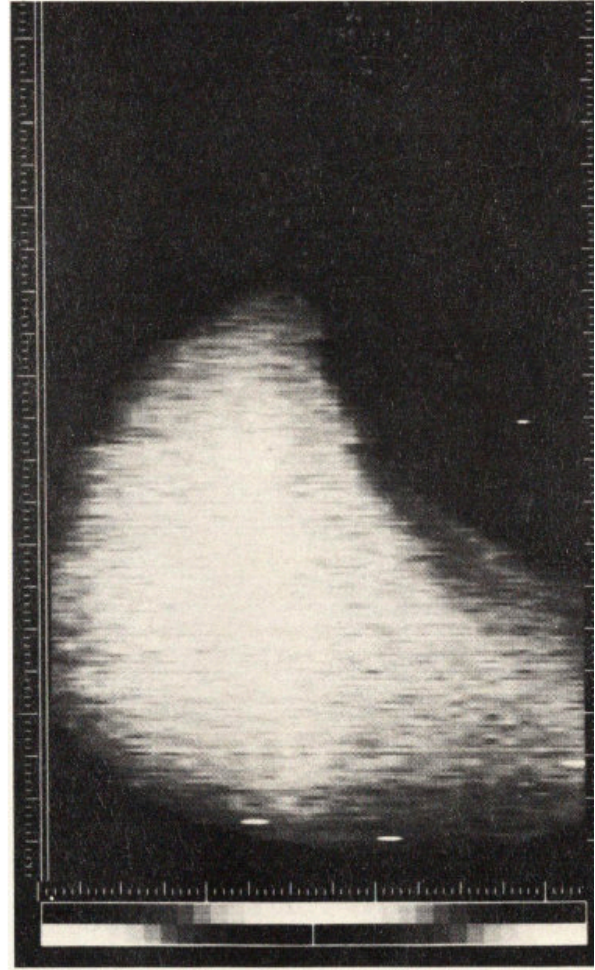
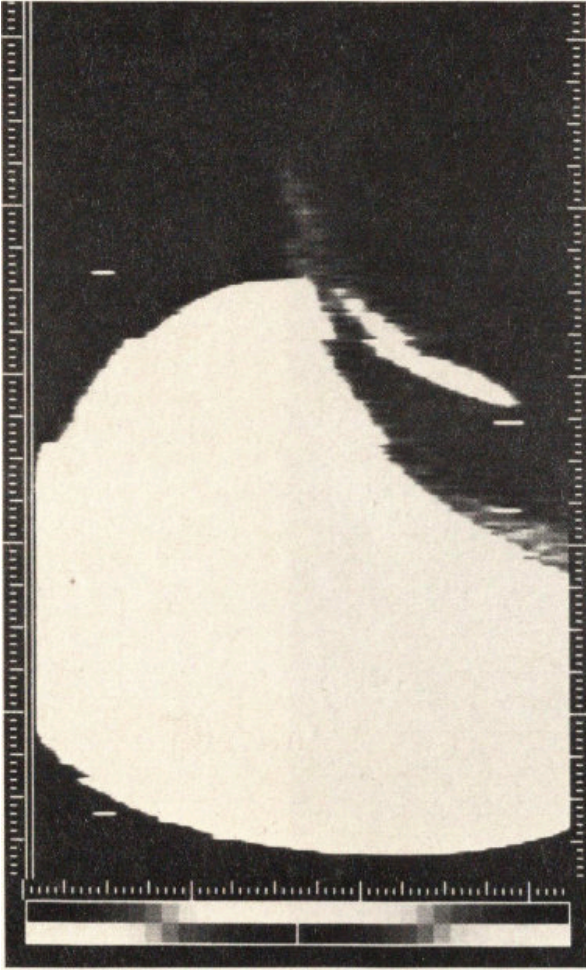
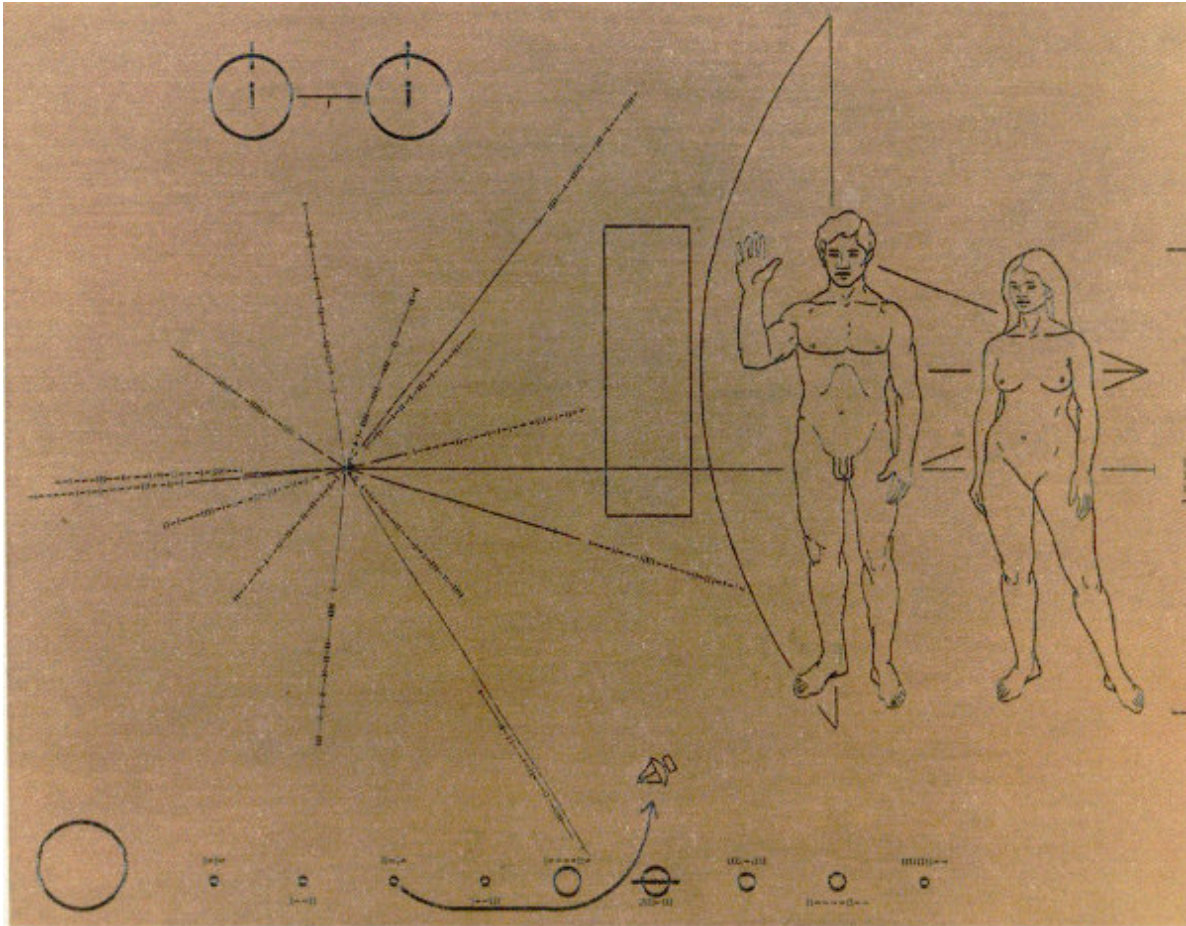


Diagram of Saturn's inner trapped radiation belts. The energetic particle fluxes generally become more intense closer to Saturn. Decreases in particle flux at the locations of Saturn's moons are due to the sweeping up of the energetic particles caused by particles striking the moons and being absorbed by them. Also, there are decreases in particle flux at the outer edge of the rings, where the energetic particles are also absorbed, so that a region free of trapped radiation is created from the outer edge of the rings to Saturn.



Infrared radiometer image of Saturn and its rings. Brightness in the image is related to temperature, with the brightest areas at about 100 K (about -280°F). The left version shows contrast in the colder regions (the rings). The right version shows contrast in the warmer regions (the planet). The image contains many separate scans from top to bottom. Each scan is displaced to the right from the one before by the motion of the spacecraft. The spacecraft was below the ring plane during most of the 3-hour observation period and was much closer to the planet at the end of the period (right of image) than at the start (left of image). Thus, the images are quite distorted. The small-scale pattern is instrument noise. In the left image, the warm infrared radiation from the planet is seen through Cassini's Division between the A- and B-rings. The brightness level in the right image implies that Saturn emits heat at a rate that is 2.5 times the rate that it absorbs energy from the sun.

OUTWARD BOUND



After increasing our knowledge of the Saturnian system manyfold, the spacecraft is now outward bound. It carries a message, engraved on a plaque, from Earth to any inhabitants of another star system who might discover the spacecraft. With its journey far from over, Pioneer Saturn travels on—to the outer reaches of space.

The Pioneer Project is managed by the Ames Research Center for the National Aeronautics and Space Administration. The spacecraft was built by TRW Systems, Redondo Beach, California.

National Aeronautics and Space Administration

Ames Research Center
Moffett Field, California 94035

Official Business

Penalty for Private Use, \$300

Postage and Fees Paid

National Aeronautics and Space Administration

NASA-451



Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Silently corrected a few palpable typos.
- Added a Table of Contents.
- In the text versions only, text in italics is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK PIONEER
SATURN ENCOUNTER ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.